

# INTELLIGENCE IS NOT GLOBAL OPTIMIZATION

*Why Gradient Descent Fails in Real-Time Physical Control*

**The Core Premise:** Most modern AI relies on global backpropagation, which is computationally heavy and biologically implausible. We demonstrate that robust physical intelligence emerges faster from local, asynchronous prediction—unlocking AI that learns in real-time on the edge.

## THE ARCHITECTURE SHIFT

| Current Deep Learning      | → | Physical-Native AI       |
|----------------------------|---|--------------------------|
| × Global Loss Optimization | → | ✓ Local Prediction Error |
| × Discrete Time Batches    | → | ✓ Continuous Time (LNN)  |
| × Dense Matrix Activations | → | ✓ Sparse Spikes (SNN)    |
| × Catastrophic Forgetting  | → | ✓ Topological Adaptation |

## THE EVIDENCE & THE GAP

- **Biology:** The brain operates on ~20W with ZERO backpropagation, using local plasticity (STDP) for real-time adaptation.
- **Physics:** Control systems in nature (e.g., insect flight) solve non-linear dynamics instantly without “training epochs.”
- **The Gap:** Standard RL (e.g., PPO, SAC) cannot adapt to sudden hardware damage (e.g., broken motor) without costly retraining.

## COMPUTATIONAL ADVANTAGE

- **O(1) Update Complexity:** Learning happens instantly at the synapse level (STDP), avoiding the  $O(N)$  backward pass of backprop.
- **Extreme Sparsity:** Unlike Transformers (dense matrix math), our network is idle until a significant physical event occurs, reducing theoretical FLOPs by orders of magnitude.

## MINIMAL VERIFICATION EXPERIMENT (MVE)

- **Task:** Control an inverted pendulum subjected to **OOD (Out-of-Distribution)** parameter jumps (e.g., 2x gravity change).
- **Agent:** Liquid-spiking network (LNN + SNN) with local plasticity. **Strictly no backprop.**
- **Baselines:**
  1. *Classical:* MPC (Heavy compute), PID (Limited adaptability).
  2. *Modern AI:* Meta-RL (Slow inference adaptation).
- **Key Metrics:**
  1. **OOD Adaptation:** Time to re-stabilize after physics jump vs. Meta-RL.
  2. **Compute Efficiency:** FLOPs & Active Parameter count vs. small MLP.

**Goal:** Prove we can adapt faster than Meta-RL with 100x less compute.

## BEACHHEAD MARKET & THE ASK

### Primary Market: FPV Racing & Defense Drones

- High-speed drones that must survive mid-air collisions or motor failure without crashing. Latency and weight are critical constraints.

**Secondary Market:** Ultra-low power IoT sensors (preventative maintenance).

**The Ask:** Seeking \$25k validation grant for a 6-month PoC (simulators, compute, edge-boards) & intros to neuromorphic experts.

## 6-MONTH COMMERCIALIZATION & VALIDATION ROADMAP

- **Months 1–2 (BUILD):** Implement LNN+SNN agent with local STDP rules. Build a robust simulation env with randomized physics parameters.
- **Months 3–4 (BENCHMARK):** Stress-test against Meta-RL in OOD scenarios (e.g., simulate wind gusts / broken actuators). Measure adaptation speed (ms) and FLOPs.
- **Months 5–6 (DEPLOYMENT PREP):** Open-source the core library. Demonstrate the algorithm running on a constrained edge device (e.g., Jetson Nano or Raspberry Pi) as a proxy for future neuromorphic hardware.

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*Seeking early feedback and mentorship on the simulation phase.*